

HCSM-71

Gravitational Anomalies on the Finite Lattice

The Anomaly Tower and the Vanishing of the Physical Gravitational Anomaly

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October 5, 2026

HCSM White Paper Series, Paper 71
Sub-Program Q1: Quantum Gauge Sector

Abstract

We construct the gravitational anomaly sector of the Hodge Complex Standard Model (HCSM) directly on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on each form degree, and the $\mathbb{Z}_2$ grading

$$\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$$

anticommutes with D . We prove the *Non-Renormalization Theorem*: $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f , exact from $\{\gamma_5, D\} = 0$. We prove the *Anomaly Tower Theorem*: on the 56-dimensional Hodge sector at $\lambda = 4$, the γ_5 -weighted trace of the form-degree operator K satisfies $\text{Tr}_{V_{56}}(\gamma_5 K^n) = 14 \cdot 2^n - 28$ for every $n \geq 1$. We prove the *Anomaly-Class-Partition Match*: at $d = 4$, the ratio $\text{Tr}(\gamma_5 K^2)/\text{Tr}(K^2) = 28/84 = 1/3$ coincides with the class-partition residual $A = 1/(d-1) = 1/3$ derived independently in HCSM-69 and HCSM-70. We prove the *Physical Gravitational Anomaly Vanishes*: $A_{\text{grav}}^{\text{phys}} = \sum_i \gamma_5^{(i)} k_i = 0$, because all 14 physical modes at $\lambda = 4$ live in Ω^0 . We prove the *Anomaly Inflow Factorization*: $A_{\text{grav}}(V_{56}) = \dim V_{14} \cdot \text{Tr}_{\ker D}(\gamma_5 K^2) = 14 \cdot 2 = 28$, extending the Tensor Product Theorem of HCSM-61. We prove the *absence of global anomalies*: the Witten $\text{SU}(2)$, π_4 , and gravitational global anomalies all vanish. The framework has zero empirical anchors in the gravitational anomaly sector. Six open items are catalogued and assigned to downstream papers in the Q1 sub-program.

Keywords: Hodge Complex Standard Model; HCSM; gravitational anomaly; anomaly tower; non-renormalization; class-partition residual; anomaly inflow; finite lattice; Q1 sub-program.

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1 Introduction

1.1 The gravitational anomaly problem

A quantum field theory with chiral fermions coupled to gravity is consistent only if its gravitational anomaly coefficients vanish. In the Standard Model, the pure gravitational anomaly cancels because the fermion content is vector-like with respect to gravity: the number of left-handed and right-handed Lorentz representations is equal at every mass dimension. The mixed gauge-gravitational anomaly cancels because the hypercharges sum to zero over the Weyl fermions.

In the Hodge Complex Standard Model (HCSM), the gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ is derived as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet at $\lambda = 4$ [1, 5]. The chiral anomaly sector was constructed in HCSM-69 [18]. The gauge anomaly sector was constructed in HCSM-70 [19]. The present paper constructs the gravitational anomaly sector.

The paper is organized around a structural motif that recurs across the three anomaly sectors of the framework. The motif is that the substrate’s native answer is zero, and the physical content is the residual that survives the cancellation. This motif is not introduced here as a postulate; it is a recurring pattern that appears independently in the chiral sector (HCSM-69), the gauge sector (HCSM-70), and, as established in the present paper, the gravitational sector.

1.2 The zero-point structure in the gravitational sector

In the gravitational anomaly sector, the zero-point structure appears in four places.

- (1) The γ_5 -weighted trace of the Hodge-Dirac operator vanishes identically: $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f (Theorem 3.1).
- (2) The gravitational anomaly on the physical 14-mode sector vanishes because all 14 modes live in Ω^0 (Theorem 6.1).
- (3) The mixed gauge-gravitational anomaly vanishes because $\text{Tr}(Y) = 0$ over the Weyl fermions [6].
- (4) The global anomalies (Witten SU(2), π_4 , gravitational) vanish (Theorems 8.1, 8.2, 8.3).

The physical content of the gravitational anomaly sector is the residual that survives the cancellation. This residual is the anomaly tower $\text{Tr}_{V_{56}}(\gamma_5 K^n) = 14 \cdot 2^n - 28$ (Theorem 4.1), which at $n = 2$ gives the coefficient 28 and at every n exhibits the binary structure 2^n of the Hodge complex.

1.3 Main results

- (1) **Non-Renormalization Theorem** (Theorem 3.1). $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f .
- (2) **Anomaly Tower Theorem** (Theorem 4.1). $\text{Tr}_{V_{56}}(\gamma_5 K^n) = 14 \cdot 2^n - 28$ for every $n \geq 1$.
- (3) **Anomaly-Class-Partition Match** (Theorem 5.1). At $d = 4$, $\text{Tr}(\gamma_5 K^2)/\text{Tr}(K^2) = 1/3 = A$.
- (4) **Physical Gravitational Anomaly Vanishes** (Theorem 6.1). $A_{\text{grav}}^{\text{phys}} = 0$.
- (5) **Anomaly Inflow Factorization** (Theorem 7.1). $A_{\text{grav}}(V_{56}) = 14 \cdot 2 = 28$.
- (6) **Absence of Global Anomalies** (Theorems 8.1–8.3).

1.4 Cross-references

The paper depends on the following HCSM companions: HCSM-00 [1] (foundations); HCSM-02 [2] (Hodge complex); HCSM-05 [4] (D_4 representation theory); HCSM-07 [6] (hypercharge quantization); HCSM-45 [8] (graviton as self-paired curvature mode); HCSM-46 [9] (graviton exact masslessness); HCSM-47 [10] (graviton self-coupling from D_4); HCSM-48 [11] (Faddeev-Popov G_N normalization); HCSM-49 [12] (Newton constant 8π factor); HCSM-50 [13] (composite-operator effective action); HCSM-61 [14] (Tensor Product Theorem); HCSM-65 [15] (Dirac equation from the Hodge-Dirac operator); HCSM-67 [16] (BRST on the finite lattice); HCSM-68 [17] (Faddeev-Popov from the D_4 commutant); HCSM-69 [18] (quantum chiral anomalies); HCSM-70 [19] (quantum gauge anomalies).

1.5 Organization

Section 2 states the substrate, the Hodge complex, the $\mathbb{Z}_2$ grading, and the form-degree operator K . Section 3 proves the Non-Renormalization Theorem. Section 4 proves the Anomaly Tower Theorem. Section 5 states the Anomaly-Class-Partition Match. Section 6 proves the vanishing of the physical gravitational anomaly. Section 7 derives the anomaly inflow factorization. Section 8 proves the absence of global anomalies. Section 9 catalogs the open items. Section 10 lists falsifiable predictions. Section 11 concludes. Appendix A gives the numerical verification. Appendix A gives web-crawler metadata.

2 Setup

2.1 The substrate and the Hodge complex

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = L^2 = 64, \quad (1)$$

selected by the unified substrate selection theorem [1]. The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16. \quad (2)$$

The Hodge complex of differential forms on Λ consists of three graded pieces

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (3)$$

of total dimension 256. The exterior derivative d and codifferential d^* satisfy $d_1 d_0 = 0$ exactly. The Hodge-Dirac operator

$$D = d + d^* \quad (4)$$

satisfies

$$D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}, \quad k = 0, 1, 2, \quad (5)$$

where Δ is the discrete 5-point Laplacian. D is self-adjoint. The kernel of D has dimension

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4, \quad (6)$$

the Betti numbers of the two-torus. This is the emergent spacetime dimension $d = \dim \ker D = 4$ [2].

2.2 The $\mathbb{Z}_2$ grading and the form-degree operator K

The $\mathbb{Z}_2$ grading is

$$\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}, \quad (7)$$

acting as $+1$ on even forms ($\Omega^0 \oplus \Omega^2$) and -1 on odd forms (Ω^1). It satisfies $\gamma_5^2 = I$ and

$$\{\gamma_5, D\} = 0. \quad (8)$$

The form-degree operator K is the diagonal operator

$$K|_{\Omega^k} = k \cdot \text{id}_{\Omega^k}, \quad k = 0, 1, 2. \quad (9)$$

It satisfies $[K, D^2] = 0$ exactly, because $D^2 = \Delta$ acts as a scalar on each form degree. It anticommutes with γ_5 in the sense

$$\gamma_5 K \gamma_5 = -K. \quad (10)$$

The physical motivation for identifying K , rather than the naive $\text{SO}(2)$ rotation generator J , with the discrete Lorentz representation weight is as follows. On a d -dimensional manifold, a differential k -form transforms under the Lorentz group as the k -th antisymmetric tensor power of the tangent representation. The form degree k is therefore the representation-theoretic label of the Lorentz weight. On the lattice, K is the diagonal operator that measures this label. The naive generator J acts only on the two components of Ω^1 at each site and does not distinguish the form degrees globally; it is not the correct global object. The distinction is verified numerically: $\text{Tr}(\gamma_5 J^2) = -1$ whereas $\text{Tr}(\gamma_5 K^2) = 28$. An independent derivation of this identification, in particular a construction of K as the lattice analogue of the continuum Lorentz generator $\sigma^{\mu\nu}$, is catalogued as Open Item 9.2 and assigned to HCSM-72.

2.3 The 56-dimensional Hodge sector at $\lambda = 4$

The discrete 5-point Laplacian has a self-paired eigenvalue $\lambda = 4$ of multiplicity $2(L - 1) = 14$ on Ω^0 [3]. The Hodge complex at $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (11)$$

of total dimension

$$\dim V_{56} = 14 + 28 + 14 = 56. \quad (12)$$

The D_4 isotypic decomposition of V_{56} is

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E, \quad (13)$$

with dimension check $9 + 5 + 7 + 7 + 28 = 56$ [2, 4].

The form-degree multiplicities are

$$\dim \Omega^0|_{\lambda=4} = 14, \quad \dim \Omega^1|_{\lambda=4} = 28, \quad \dim \Omega^2|_{\lambda=4} = 14. \quad (14)$$

These are theorems of the torus, verified at machine precision in Appendix A.

3 The Non-Renormalization Theorem

3.1 The substrate identity

From (8), the Hodge-Dirac operator anticommutes with the $\mathbb{Z}_2$ grading:

$$\gamma_5 D \gamma_5^{-1} = -D. \quad (15)$$

Therefore

$$\gamma_5 D^{2n} \gamma_5^{-1} = D^{2n}, \quad n \geq 0. \quad (16)$$

3.2 Vanishing of the regulated trace

Theorem 3.1 (Non-Renormalization). *For every even function f , the regulated trace vanishes:*

$$\text{Tr}(\gamma_5 f(D)) = 0. \quad (17)$$

Proof. For an even function f , the operator identity $\gamma_5 D^{2n} \gamma_5^{-1} = D^{2n}$ implies

$$\gamma_5 f(D) \gamma_5^{-1} = f(D). \quad (18)$$

The trace of γ_5 on the full 256-dimensional Hodge complex vanishes:

$$\text{Tr}(\gamma_5) = 128 \cdot (+1) + 128 \cdot (-1) = 0. \quad (19)$$

Each D^{2n} preserves the form-degree grading, so the trace

$$\text{Tr}(\gamma_5 D^{2n}) = \sum_{k=0}^2 \gamma_5(k) \text{Tr}_{\Omega^k}(D^{2n}) = \sum_{\lambda} \gamma_5(\lambda) \lambda^n = 0 \quad (20)$$

vanishes by the spectral symmetry: for every eigenmode of D^2 with form-parity $+1$ and eigenvalue λ , there is a corresponding eigenmode with form-parity -1 and eigenvalue λ [2, 18]. Therefore $\text{Tr}(\gamma_5 f(D)) = 0$. $\square$

3.3 Verification over nine regulators

The identity $\text{Tr}(\gamma_5 f(D)) = 0$ is verified numerically in Appendix A for nine even regulators: $\exp(-D^2)$, $\exp(-D^2/2)$, $1/(1+D^2)$, $\exp(-D^4)$, $\cos(D)$, $\text{sinc}(D)$, $\exp(-D^2/4)$, $1/(2+D^2)$, and $\exp(-|D|)$. All nine traces vanish to machine precision (residual $< 10^{-14}$).

3.4 Consequence

Corollary 3.2. *There is no gravitational anomaly in the free theory on the Hodge complex at any loop order.*

Proof. The anomaly coefficient is the regulated trace $\text{Tr}(\gamma_5 f(D))$ for some even regulator f . By Theorem 3.1, this vanishes identically. The statement holds at every order because the identity $\{\gamma_5, D\} = 0$ is exact, not perturbative. $\square$

4 The Anomaly Tower

4.1 The spectral decomposition of V_{56}

From the multiplicities (14), the form-degree operator K on V_{56} has three distinct eigenvalues: 0 with multiplicity 14, 1 with multiplicity 28, and 2 with multiplicity 14. The $\mathbb{Z}_2$ grading is $\gamma_5 = +1$ on the $k = 0$ and $k = 2$ sectors, and $\gamma_5 = -1$ on the $k = 1$ sector.

4.2 The closed form

Theorem 4.1 (Anomaly Tower). *On the 56-dimensional Hodge sector at $\lambda = 4$, the γ_5 -weighted trace of the form-degree operator K satisfies*

$$\text{Tr}_{V_{56}}(\gamma_5 K^n) = 14 \cdot 2^n - 28, \quad n \geq 1. \quad (21)$$

Equivalently,

$$\text{Tr}_{V_{56}}(\gamma_5 K^n) = 28(2^{n-1} - 1), \quad n \geq 2. \quad (22)$$

Proof. By the spectral decomposition of K on V_{56} ,

$$\text{Tr}_{V_{56}}(\gamma_5 K^n) = \sum_{k=0}^2 \gamma_5(k) \cdot \dim(\Omega^k|_{\lambda=4}) \cdot k^n. \quad (23)$$

Substituting $\gamma_5 = (+1, -1, +1)$ and the multiplicities (14):

$$\text{Tr}_{V_{56}}(\gamma_5 K^n) = (+1) \cdot 14 \cdot 0^n + (-1) \cdot 28 \cdot 1^n + (+1) \cdot 14 \cdot 2^n. \quad (24)$$

For $n \geq 1$, the first term vanishes ($0^n = 0$), giving

$$\text{Tr}_{V_{56}}(\gamma_5 K^n) = -28 + 14 \cdot 2^n = 14 \cdot 2^n - 28. \quad (25)$$

The factored form $28(2^{n-1} - 1)$ follows by algebra. $\square$

4.3 The binary structure

Theorem 4.1 exhibits the anomaly sector's binary structure. Writing $\text{Tr}_{V_{56}}(\gamma_5 K^n) = 28(2^{n-1} - 1)$ for $n \geq 2$, the coefficient 28 is exactly the dimension of $\Omega^1|_{\lambda=4}$, and the factor $2^{n-1} - 1$ is a Mersenne-type binary growth determined by the form-degree ladder.

The first three nontrivial values are

$$\text{Tr}_{V_{56}}(\gamma_5 K^2) = 28, \quad \text{Tr}_{V_{56}}(\gamma_5 K^3) = 84, \quad \text{Tr}_{V_{56}}(\gamma_5 K^4) = 196. \quad (26)$$

4.4 The trace $\text{Tr}(K^n)$

The unweighted trace is

$$\text{Tr}_{V_{56}}(K^n) = 14 \cdot 0^n + 28 \cdot 1^n + 14 \cdot 2^n. \quad (27)$$

For $n \geq 1$, this is $\text{Tr}_{V_{56}}(K^n) = 28 + 14 \cdot 2^n$.

Proposition 4.2. *The traces satisfy*

$$\text{Tr}_{V_{56}}(K^n) = 28 + 14 \cdot 2^n, \quad \text{Tr}_{V_{56}}(\gamma_5 K^n) = -28 + 14 \cdot 2^n, \quad n \geq 1. \quad (28)$$

4.5 The odd-derivative selection rule (extended)

HCSM-47 established an odd-derivative selection rule for the A_1 -invariant content of the graviton vertex spaces: the content vanishes for odd numbers of derivatives [10]. The present result is not in conflict with HCSM-47. HCSM-47 concerns the A_1 -invariant content of $\mathcal{V}_{n,m}$, while the present result concerns the γ_5 -weighted trace $\text{Tr}_{V_{56}}(\gamma_5 K^n)$. The γ_5 weighting changes the parity of the trace: the odd- n values $\text{Tr}_{V_{56}}(\gamma_5 K^{2m+1})$ are generally nonzero.

5 The Anomaly-Class-Partition Match

5.1 The class-partition residual

HCSM-69 defines the class-partition residual as

$$A = \frac{\text{Tr}[Q_5^3]}{d \cdot |A|} = \frac{1}{d-1} = \frac{1}{3} \text{ at } d=4, \quad (29)$$

where Q_5 is the chiral charge and $|A| = d-1$ is the color-block size [18]. The residual has the F-defect interpretation: the orientation flip F has exactly one nonzero- Q_5 pair $\{\text{dark}, \tau\}$, and $A = 1/|A|$ is the single F-defect charge divided by the color-block size.

HCSM-70 extends the residual to the gauge anomaly sector [19].

5.2 The gravitational ratio

Theorem 5.1 (Anomaly-Class-Partition Match). *At $d = 4$, the gravitational anomaly ratio satisfies*

$$\frac{\text{Tr}_{V_{56}}(\gamma_5 K^2)}{\text{Tr}_{V_{56}}(K^2)} = \frac{28}{84} = \frac{1}{3}. \quad (30)$$

This value coincides with the class-partition residual $A = 1/(d-1) = 1/3$ of (29).

Proof. By Theorem 4.1, $\text{Tr}_{V_{56}}(\gamma_5 K^2) = 14 \cdot 4 - 28 = 28$. By Proposition ??, $\text{Tr}_{V_{56}}(K^2) = 28 + 14 \cdot 4 = 84$. The ratio is $28/84 = 1/3$. At $d = 4$, the class-partition residual (29) is $1/(d-1) = 1/3$. $\square$

5.3 Scope of the match

Theorem 5.1 establishes a numerical match at $d = 4$ between the gravitational ratio $\text{Tr}(\gamma_5 K^2)/\text{Tr}(K^2)$ and the class-partition residual $A = 1/(d-1)$. The two quantities are computed independently: the ratio from the Hodge complex of the present paper, the residual from the class partition of HCSM-69 and HCSM-70.

Whether the two $1/3$ values are structurally identical for all d , or numerically coincident at $d = 4$, is not resolved in this paper. A general- d computation of the gravitational ratio, to be compared with $1/(d-1)$, is catalogued as Open Item 9.1 and assigned to HCSM-72.

6 The Physical Gravitational Anomaly Vanishes

6.1 The 14 physical modes and the form degree

The 14 modes at $\lambda = 4$ are identified with one Standard Model generation via the Midpoint Sorting Theorem [7]. All 14 modes live in $\Omega^0|_{\lambda=4}$, and therefore have form degree $k = 0$.

Theorem 6.1 (Physical Gravitational Anomaly Vanishes). *Given the emergent Lorentz structure of signature $(3, 1)$ constructed in HCSM-61 and HCSM-65 [14, 15], the physical gravitational anomaly coefficient is*

$$A_{\text{grav}}^{\text{phys}} = \sum_{i=1}^{14} \gamma_5^{(i)} k_i = 0, \quad (31)$$

because all 14 physical modes are Lorentz scalars in the Ω^0 sector.

Proof. The physical gravitational anomaly is $\text{Tr}_{V_{14}}(\gamma_5 \sigma^{\mu\nu} \sigma^{\rho\sigma})$, with $\sigma^{\mu\nu} = (i/2)[\gamma^\mu, \gamma^\nu]$ the emergent 4D Lorentz generators from HCSM-61 and HCSM-65. On the physical sector $V_{14} \subset \Omega^0|_{\lambda=4}$, the form degree is $k = 0$, so the Lorentz weight vanishes and $\sigma^{\mu\nu}$ acts trivially. Therefore

$$A_{\text{grav}}^{\text{phys}} = \sum_{i=1}^{14} \gamma_5^{(i)} \cdot 0 = 0. \quad (32)$$

$\square$

Remark 6.2. The vanishing of the physical gravitational anomaly is the discrete analogue of the continuum statement that the Standard Model fermion content is vector-like with respect to gravity. In HCSM, the vanishing is structural: it follows from the form-degree grading of the Hodge complex, which places all physical modes in the $k = 0$ sector.

6.2 The mixed gauge-gravitational anomaly

The mixed gauge-gravitational anomaly coefficient is

$$A^{Y-\text{grav}} = \text{Tr}_{V_{56}}(Y \cdot \gamma_5 \cdot \sigma^{\mu\nu} \sigma^{\rho\sigma}). \quad (33)$$

By the Tensor Product Theorem $V_{56} \cong V_{14} \otimes \ker D$ [14], the trace factorizes:

$$A^{Y-\text{grav}} = \text{Tr}_{V_{14}}(Y \cdot \gamma_5) \cdot \text{Tr}_{\ker D}(\sigma^{\mu\nu} \sigma^{\rho\sigma}). \quad (34)$$

The first factor is $\text{Tr}_{V_{14}}(Y \gamma_5) = \sum_i Y_i = 0$, by the Standard Model anomaly cancellation conditions derived in HCSM-07 [6]. Therefore $A^{Y-\text{grav}} = 0$.

7 Anomaly Inflow

7.1 The Tensor Product Theorem

The Tensor Product Theorem of HCSM-61 states

$$V_{56} \cong V_{14} \otimes \ker D \quad (35)$$

as D_4 representations [14]. The internal space is V_{14} (the 14 modes at $\lambda = 4$ carrying the Standard Model generation content); the spacetime tangent space is $\ker D$ (the 4-dimensional kernel of the Hodge-Dirac operator, with Lorentzian metric of signature $(3, 1)$).

7.2 Anomaly inflow factorization

Theorem 7.1 (Anomaly Inflow Factorization). *The gravitational anomaly on V_{56} factorizes as*

$$A_{\text{grav}}(V_{56}) = \dim V_{14} \cdot \text{Tr}_{\ker D}(\gamma_5 K^2) = 14 \cdot 2 = 28. \quad (36)$$

Proof. By the tensor product structure, the γ_5 -weighted trace factorizes:

$$\text{Tr}_{V_{56}}(\gamma_5 K^2) = \text{Tr}_{V_{14}}(1) \cdot \text{Tr}_{\ker D}(\gamma_5 K^2). \quad (37)$$

The first factor is $\text{Tr}_{V_{14}}(1) = 14$. For the second factor, $\ker D$ has Betti numbers $b_0 = 1$, $b_1 = 2$, $b_2 = 1$; the grading is $\gamma_5 = +1$ on b_0 and b_2 , and -1 on b_1 ; the form-degree operator satisfies $K = 0$ on b_0 , $K = 1$ on b_1 , $K = 2$ on b_2 . Therefore

$$\text{Tr}_{\ker D}(\gamma_5 K^2) = (+1) \cdot 1 \cdot 0^2 + (-1) \cdot 2 \cdot 1^2 + (+1) \cdot 1 \cdot 2^2 = 0 - 2 + 4 = 2. \quad (38)$$

The product is $14 \cdot 2 = 28$. $\square$

7.3 Consistency

Theorem 7.1 reproduces the direct computation of $\text{Tr}_{V_{56}}(\gamma_5 K^2) = 28$ from Theorem 4.1 at $n = 2$. The factorization $56 = 14 \cdot 4$ and the evaluation $\text{Tr}_{\ker D}(\gamma_5 K^2) = 2$ on the Betti numbers $(1, 2, 1)$ are verified at machine precision in Appendix A.

8 Global Anomalies

8.1 The Witten SU(2) global anomaly

Theorem 8.1 (Witten Anomaly Evasion). *The Witten global SU(2) anomaly is evaded: the number of $\text{SU}(2)_L$ doublets is even.*

Proof. The Witten anomaly is $\mathbb{Z}_2$ -valued: $\pi_4(\text{SU}(2)) = \mathbb{Z}_2$. The anomaly is absent if the number of $\text{SU}(2)_L$ doublets is even. On the 14-mode multiplet, the number of $\text{SU}(2)_L$ doublets is 4 (three colors of Q_L plus one lepton doublet L_L ; see HCSM-70 [19] Theorem 9.2). Since 4 is even, the Witten anomaly is evaded. $\square$

8.2 The π_4 anomaly

Theorem 8.2 (π_4 Anomaly Vanishes). *The π_4 anomaly vanishes on the 14-mode multiplet:*

$$\pi_4(\mathrm{SU}(2)_L) = 0. \quad (39)$$

Proof. The π_4 anomaly is $\mathbb{Z}_2$ -valued. It vanishes if the number of fermions in the fundamental representation is even. On the 14-mode multiplet, the number of fundamental fermions is 4 (even). Therefore the π_4 anomaly vanishes. $\square$

8.3 The gravitational global anomaly

Theorem 8.3 (Gravitational Global Anomaly Vanishes). *The gravitational global anomaly vanishes on the 14-mode multiplet.*

Proof. By Theorem 6.1, $A_{\mathrm{grav}}^{\mathrm{phys}} = 0$. The corresponding global anomaly is the integral of the anomaly density over the compactified spacetime, which vanishes identically because the density vanishes pointwise on the physical sector. $\square$

8.4 Summary

Theorem 8.4 (Absence of Global Anomalies). *All global anomalies vanish on the 14-mode multiplet:*

- (1) *The Witten global $\mathrm{SU}(2)$ anomaly is evaded (Theorem 8.1).*
- (2) *The π_4 anomaly vanishes (Theorem 8.2).*
- (3) *The gravitational global anomaly vanishes (Theorem 8.3).*

9 Open Items

The following open items are catalogued for assignment to downstream papers in the $Q1$ sub-program.

Open Item 9.1 (Structural identity of the $1/3$ match). Theorem 5.1 establishes a numerical match at $d = 4$ between the gravitational ratio $\mathrm{Tr}(\gamma_5 K^2)/\mathrm{Tr}(K^2)$ and the class-partition residual $A = 1/(d - 1)$. Whether the two $1/3$ values are structurally identical for all d , or numerically coincident at $d = 4$, is not resolved in this paper. A general- d computation is assigned to HCSM-72.

Open Item 9.2 (K as the discrete Lorentz generator). The form-degree operator K is identified with the discrete Lorentz weight because each form degree transforms as the k -th antisymmetric power of the tangent representation. An independent derivation of this identification, distinguishing K from the naive $\mathrm{SO}(2)$ generator J (which gives $\mathrm{Tr}(\gamma_5 J^2) = -1$ rather than 28), is assigned to HCSM-72.

Open Item 9.3 (4D Lorentz structure). The physical gravitational anomaly coefficient uses the emergent 4D Lorentz structure $\sigma^{\mu\nu} = (i/2)[\gamma^\mu, \gamma^\nu]$ constructed in HCSM-61 and HCSM-65. The present paper imports this structure; an explicit construction of $\sigma^{\mu\nu}$ on the Hodge complex is not repeated here.

Open Item 9.4 (Weyl hypercharges). The mixed gauge-gravitational anomaly coefficient uses the vanishing of $\sum_i Y_i$ over the Weyl fermions. The Weyl-hypercharge assignment is derived in HCSM-07 and imported here.

Open Item 9.5 (BRST operator for Wess-Zumino). The Wess-Zumino consistency condition $s\mathcal{A} = 0$ uses the BRST operator s constructed in HCSM-67 and HCSM-68. The present paper invokes the commuting of K with D^2 as a sufficient condition for consistency but does not rederive the BRST structure.

Open Item 9.6 (Anomaly polynomial I_8). The anomaly polynomial I_8 on the Hodge complex is a restatement of the Non-Renormalization Theorem 3.1. An explicit construction of I_8 from the gauge field strength F and the Riemann tensor R on the lattice is not given here.

10 Falsifiable Predictions

- (1) **Non-renormalization.** $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f . Any confirmed nonzero lattice anomaly at any scale would falsify Theorem 3.1.
- (2) **Anomaly tower.** $\text{Tr}_{V_{56}}(\gamma_5 K^n) = 14 \cdot 2^n - 28$ for every $n \geq 1$. A different trace value would falsify Theorem 4.1.
- (3) **Gravitational ratio.** $\text{Tr}(\gamma_5 K^2)/\text{Tr}(K^2) = 1/3$ at $d = 4$. A different ratio would falsify Theorem 5.1.
- (4) **Physical anomaly.** $A_{\text{grav}}^{\text{phys}} = 0$. A nonzero physical gravitational anomaly would falsify Theorem 6.1.
- (5) **Inflow.** $A_{\text{grav}}(V_{56}) = 28$. A different value would falsify Theorem 7.1.
- (6) **Global anomalies.** Witten $\text{SU}(2)$, π_4 , gravitational all vanish. A confirmed nonzero global anomaly would falsify Theorem 8.4.
- (7) **No R^2 corrections.** By HCSM-50, the graviton effective action is exactly Einstein-Hilbert [13]. Any confirmed R^2 or Gauss-Bonnet correction would falsify the composite-operator program.

11 Conclusion

We have constructed the gravitational anomaly sector of the Hodge Complex Standard Model on the finite substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.

The Non-Renormalization Theorem establishes that $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f , exact from $\{\gamma_5, D\} = 0$. This is the substrate's native answer at the gravitational anomaly level.

The Anomaly Tower Theorem establishes that $\text{Tr}_{V_{56}}(\gamma_5 K^n) = 14 \cdot 2^n - 28$ for every $n \geq 1$. The coefficient 28 is the dimension of the $\Omega^1|_{\lambda=4}$ sector, and the factor 2^n is the binary growth of the form-degree ladder. The first nontrivial value $\text{Tr}(\gamma_5 K^2) = 28$ is the anomaly coefficient; its ratio to $\text{Tr}(K^2) = 84$ is $1/3$.

The Anomaly-Class-Partition Match establishes that at $d = 4$, $\text{Tr}(\gamma_5 K^2)/\text{Tr}(K^2) = 1/3$ coincides with the class-partition residual $A = 1/(d-1) = 1/3$ derived independently in HCSM-69 and HCSM-70. The two $1/3$ values are computed independently; whether they are structurally identical for all d or numerically coincident at $d = 4$ is catalogued as Open Item 9.1 and assigned to HCSM-72.

The Physical Gravitational Anomaly vanishes because all 14 physical modes live in Ω^0 and therefore have form degree $k = 0$.

The Anomaly Inflow Factorization $\text{Tr}_{V_{56}}(\gamma_5 K^2) = 14 \cdot 2 = 28$ extends the Tensor Product Theorem of HCSM-61 to the gravitational anomaly sector.

All global anomalies (Witten $\text{SU}(2)$, π_4 , gravitational) vanish.

Six open items are catalogued and assigned to downstream papers: the structural identity of the 1/3 match (HCSM-72); the derivation of K as the discrete Lorentz generator (HCSM-72); the explicit 4D Lorentz structure (HCSM-61, HCSM-65); the Weyl-hypercharge assignment (HCSM-07); the BRST operator for Wess-Zumino consistency (HCSM-67, HCSM-68); and the explicit anomaly polynomial I_8 .

The framework has zero empirical anchors in the gravitational anomaly sector.

A Numerical Verification

All numerical claims are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-71_verification.py` constructs the substrate, the Hodge complex, and the V_{56} sector at $\lambda = 4$, and verifies the following identities.

A.1 Substrate and Hodge complex

The discrete exterior derivatives d_0 and d_1 satisfy $d_1 d_0 = 0$ exactly. The Hodge-Dirac operator D is self-adjoint. The anticommutator $\{\gamma_5, D\} = 0$ exactly. The kernel of D has dimension 4, matching $b_0 + b_1 + b_2 = 1 + 2 + 1$. The smallest nonzero eigenvalue of D is $2 \sin(\pi/8) = 0.765367$.

A.2 The 56-dimensional sector at $\lambda = 4$

The multiplicity of $\lambda = 4$ in D^2 is 56. The traces on V_{56} are

$$\text{Tr}(\gamma_5) = 0, \quad (40)$$

$$\text{Tr}(K) = 56, \quad (41)$$

$$\text{Tr}(\gamma_5 K) = 0, \quad (42)$$

$$\text{Tr}(\gamma_5 K^2) = 28, \quad (43)$$

$$\text{Tr}(K^2) = 84. \quad (44)$$

A.3 Non-Renormalization over nine regulators

The regulated trace $\text{Tr}(\gamma_5 f(D))$ is verified to be zero to machine precision for the nine even regulators

$$e^{-D^2}, \quad e^{-D^2/2}, \quad \frac{1}{1+D^2}, \quad e^{-D^4}, \quad \cos(D), \quad \text{sinc}(D), \quad e^{-D^2/4}, \quad \frac{1}{2+D^2}, \quad e^{-|D|}. \quad (45)$$

The largest residual across all nine regulators is 7.88×10^{-15} .

A.4 The anomaly tower

The trace $\text{Tr}_{V_{56}}(\gamma_5 K^n)$ is computed for $n = 1, \dots, 9$ and matches the closed form $14 \cdot 2^n - 28$ to machine precision. The values are

$$0, \quad 28, \quad 84, \quad 196, \quad 420, \quad 868, \quad 1764, \quad 3556, \quad 7140. \quad (46)$$

A.5 The anomaly-class-partition match

The ratio $\text{Tr}_{V_{56}}(\gamma_5 K^2) / \text{Tr}_{V_{56}}(K^2) = 28/84 = 0.3333333333$ matches $1/(d-1) = 1/3 = 0.3333333333$ to machine precision.

A.6 The anomaly inflow

The factorization $\text{Tr}_{V_{56}}(\gamma_5 K^2) = \text{Tr}_{V_{14}}(1) \cdot \text{Tr}_{\ker D}(\gamma_5 K^2)$ is verified: $14 \cdot 2 = 28$.

A.7 The naive Lorentz generator

For contrast, the naive $SO(2)$ generator J on Ω^1 gives $\text{Tr}_{V_{56}}(\gamma_5 J^2) = -1$, in contrast to $\text{Tr}_{V_{56}}(\gamma_5 K^2) = 28$. The distinction is the numerical basis for the identification of K , rather than J , with the discrete Lorentz weight (Open Item 9.2).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

A Web Crawler Metadata

A.1 Dublin Core

- **DC.title:** HCSM-71: Gravitational Anomalies on the Finite Lattice
- **DC.creator:** Preschutti, Stanley
- **DC.creator.orcid:** 0009-0004-5445-1744
- **DC.date.issued:** 2026-10-05
- **DC.type:** Text / Preprint / White Paper
- **DC.format:** application/pdf; text/x-latex
- **DC.language:** en
- **DC.publisher:** Entropia Research Institute
- **DC.rights:** All rights reserved
- **DC.subject:** Hodge Complex Standard Model; HCSM; gravitational anomaly; anomaly tower; non-renormalization; class-partition residual; anomaly inflow; finite lattice; Q1 sub-program
- **DC.identifier:** HCSM-71-2026-10-05

A.2 BibTeX

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@article{PreschuttiHCSM71,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-71}: Gravitational Anomalies on the Finite Lattice},  
  journal = {HCSM White Paper Series},  
  year = {2026},  
  month = {October},  
  note = {ORCID: 0009-0004-5445-1744},  
  keywords = {Hodge Complex Standard Model, HCSM, gravitational anomaly,  
              anomaly tower, non-renormalization, class-partition residual,  
              anomaly inflow, finite lattice, Q1 sub-program}  
}
```

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